# 1 Model

We develop a static general equilibrium model of multinational production in which the strategic agent is the ownership group. A group can operate plants in several countries, chooses which of its plants serve each market, and takes into account when it sets quantities that its own plants compete with one another. Markups therefore depend on the group's total share of the market rather than on the share of any single plant. Profits are paid to the countries that own each group, and ownership is recorded group by group rather than country by country. Proofs are in the Appendix.

## 1.1 Model Set-Up

**Agents.** There is a finite set of countries $\mathcal{N} = \{1, \dots, N\}$ and a finite set of sectors $\mathcal{K} = \{1, \dots, K\}$. Each country $n$ is inhabited by a representative household that supplies $L_n$ units of labor inelastically and owns shares in firms. Production takes place in plants, and every plant belongs to exactly one group $g \in \mathcal{G}$, which we identify with its global ultimate parent. Each group has a headquarters country $h(g)$ and operates in a single sector $k(g)$.[1] Groups are of two types. A *parent group* $g \in \mathcal{G}^P$ is headquartered in a country of a set $\mathcal{N}^P \subseteq \mathcal{N}$ and has one potential plant in every country of $\mathcal{N}^P$, its headquarters country included. A *stand-alone firm* $g \in \mathcal{G}^S$ has a single potential plant, located in $h(g)$; countries outside $\mathcal{N}^P$ host only stand-alone firms. We index plants by $i$, write $g(i)$ and $\ell(i)$ for the group and the location of plant $i$ and $k(i) = k(g(i))$ for its sector, and let $\mathcal{P}_g$ be the set of potential plants of group $g$. Each plant produces one differentiated variety and can sell it in any country. A *market* is a destination–sector pair $(n, k)$. Whether a group is multinational is an equilibrium outcome: a group is multinational if its active plants are located in more than one country, an active plant located outside its group's headquarters country is a foreign affiliate, and the home plant of a multinational group is a domestic multinational. A stand-alone firm is never multinational.

**Representative household.** The household in country $n$ has nested CES preferences,

$$U_n = \Big[\sum_{k \in \mathcal{K}} \beta_k^{1/\eta}\, C_{nk}^{(\eta-1)/\eta}\Big]^{\eta/(\eta-1)}, \qquad C_{nk} = \Big[\sum_{i \in \mathcal{A}_{nk}} q_{i,n}^{(\sigma_k-1)/\sigma_k}\Big]^{\sigma_k/(\sigma_k-1)}, \tag{1}$$

where $\mathcal{A}_{nk}$ is the set of plants active in market $(n, k)$, $\beta_k > 0$ with $\sum_k \beta_k = 1$, and $\sigma_k > \eta \geq 1$: varieties within a sector are closer substitutes than goods in different sectors. The case $\eta = 1$ is the Cobb–Douglas limit. The associated price indices are

$$P_{nk} = \Big[\sum_{i \in \mathcal{A}_{nk}} p_{i,n}^{1-\sigma_k}\Big]^{1/(1-\sigma_k)}, \qquad P_n = \Big[\sum_k \beta_k P_{nk}^{1-\eta}\Big]^{1/(1-\eta)}. \tag{2}$$

The household spends its income $X_n$, allocates the share $\beta_k P_{nk}^{1-\eta}/P_n^{1-\eta}$ of it to sector $k$, and its welfare is real income $X_n/P_n$. Firms buy intermediate inputs as the same sectoral composites (below), so all purchases of sector-$k$ varieties in country $n$, by households and by firms, are aggregated by $C_{nk}$. Writing $q_{i,n}$ for the total quantity of plant $i$'s variety sold in $n$ and $E_{nk}$ for total spending on sector $k$ in $n$, demand is

$$q_{i,n} = p_{i,n}^{-\sigma_k}\, P_{nk}^{\sigma_k - 1} E_{nk}. \tag{3}$$

[1] This restriction is without loss for behavior. Marginal costs are constant and fixed costs are specific to each plant–market pair, so a group's decisions separate across markets (Section 1.2). A group active in several sectors is equivalent to several single-sector groups with the same owners; what the restriction rules out is a capability shared across sectors.

Market sizes $E_{nk}$ are determined in general equilibrium (Section 1.3).[2]

**Production technology.** Labor is the only primary factor. A plant in sector $k$ located in $\ell$ combines labor with a bundle of intermediate inputs. Writing $h_i \equiv h(g(i))$ for the headquarters country of plant $i$, its unit cost of production is

$$a_i = \frac{1}{\varphi_i}\, w_\ell^{1-\nu_k} \left(P_{\ell k}^{\mathrm{IO}}\right)^{\nu_k} \gamma_{h_i \ell}, \qquad P_{\ell k}^{\mathrm{IO}} = \prod_{k' \in \mathcal{K}} P_{\ell k'}^{\omega_{k'k}}, \tag{4}$$

where $w_\ell$ is the wage, $\nu_k \in [0,1)$ is the share of intermediates in costs, and $\omega_{k'k} \geq 0$, with $\sum_{k'} \omega_{k'k} = 1$, is the share of sector $k$'s intermediate spending that falls on sector $k'$ (Caliendo and Parro, 2015); the constant of the Cobb–Douglas cost function is absorbed into $\varphi_i$. The term $\gamma_{h\ell} \geq 1$, with $\gamma_{hh} = 1$, is the efficiency loss from producing away from headquarters (Ramondo and Rodríguez-Clare, 2013; Head and Mayer, 2019). Intermediates are bought in the markets $(\ell, k')$ of the country where the plant operates, which include imported varieties and are subject to that country's tariffs. At given wages, the entry of a plant into a market lowers its price index and hence the input costs of every producer located in that country; in general equilibrium the same entry also raises local labor demand, and the net effect on costs depends on both.

**Head-office capability.** The productivity of plant $i$ combines a location, a plant and a group component,

$$\varphi_i = z_\ell\, e^{u_i} \times \begin{cases} \exp\left[(1+\delta^P \alpha_k)\, x_g\right] & \text{if } g \in \mathcal{G}^P, \\ \exp\left[-\delta^S \alpha_k\right] & \text{if } g \in \mathcal{G}^S,\ h(g) \in \mathcal{N}^P, \\ 1 & \text{if } g \in \mathcal{G}^S,\ h(g) \notin \mathcal{N}^P, \end{cases} \tag{5}$$

where $z_\ell > 0$ is the productivity of location $\ell$, common to all plants there whatever their owner; $u_i$ is an idiosyncratic plant draw, normal with mean zero and independent across plants; $x_g \geq 0$ is the capability of parent group $g$, with $e^{x_g}$ Pareto with unit lower bound and shape $\vartheta$; and $\alpha_k \in [0,1]$ indexes the head-office intensity, or complexity, of sector $k$ (Antràs, 2003; Antràs and Helpman, 2004). Head-office services (management, technology, brands) are a non-rival capability of the group that all of its plants share, so they enter productivity rather than the cost of inputs. A stand-alone firm has no access to them, and its productivity falls short by a factor that grows with complexity at rate $\delta^S \geq 0$. A parent group's capability is worth more in complex sectors: in sector $k$ the group component $e^{(1+\delta^P \alpha_k) x_g}$ is Pareto with shape $\vartheta/(1+\delta^P \alpha_k)$, so the distribution of capability has a thicker tail where complexity is higher. Stand-alone firms in countries outside $\mathcal{N}^P$ stand in for all supply from those countries, multinational or not, and carry no penalty. The complexity index $\alpha_k$ appears only in (5).[3]

[2] Intermediate inputs are combined across sectors with Cobb–Douglas weights, so the upper-tier elasticity of intermediate demand is one. When intermediates are used ($\nu_k > 0$ below) we therefore set $\eta = 1$, so that final and intermediate demand share the same upper-tier elasticity and the markup in (10) is the one firms face. With $\eta = 1$, spending on each sector is a fixed share of total spending, so a market does not expand when its price index falls; the case $\eta > 1$ is used only without intermediates.

[3] The common alternative treats head-office services as a factor input bought at the parent's wage, so that the labor term in (4) becomes $(w_h^{\alpha_k} w_\ell^{1-\alpha_k})^{1-\nu_k}$ and the fixed cost (7) is paid in the same mix of labor (Antràs, 2003). A plant abroad then pays a foreign wage, which links factor markets across countries and can overturn the substitutability of labor across countries on which uniqueness of wages rests (Appendix C).

**Pool of potential groups.** The set of potential groups is drawn once, before any decision is taken, and the equilibrium is computed for a given draw. Each country–sector pair $(n, k)$ has a given number $M^S_{nk}$ of potential stand-alone firms, and each sector $k$ has a given number $M^P_k$ of potential parent groups. The headquarters country of each potential parent group is drawn independently from $\mathcal{N}^P$ with probability proportional to $L_h z_h^\zeta$, with $\zeta \geq 0$, and its capability $x_g$ is drawn from the same distribution whatever its headquarters. Larger and more productive countries therefore host more potential parents, and their best parents tend to be better only because more draws are taken there (Gaubert and Itskhoki, 2021). Because each market contains a finite number of groups, the draw of the pool matters for outcomes, and every equilibrium object below is conditional on it.

**Trade.** Shipping goods from $\ell$ to $n$ entails an iceberg cost $d_{\ell n} \geq 1$, with $d_{nn} = 1$. Trade costs are incurred from the country where the good is produced, not from headquarters, so a single plant can serve several destinations, including third markets, as an export platform (Tintelnot, 2017). Country $n$ levies a tariff $t_{\ell nk} > -1$ on sector-$k$ goods produced in $\ell$, with $t_{nnk} = 0$; negative values are import subsidies. Tariffs depend on where a good is produced, not on who owns the plant that produces it, so a country's tariffs also fall on the foreign affiliates of the groups its residents own. We model the tariff as a proportional wedge on the cost of delivering the good. The delivered marginal cost of plant $i$ in market $(n, k)$ is

$$c_{i,n} = a_i \, d_{\ell n} \, (1 + t_{\ell nk}), \qquad \ell = \ell(i), \tag{6}$$

and country $n$ collects $t_{\ell nk} \, a_i d_{\ell n} q_{i,n}$ on the sales of plant $i$.[4]

**Multinational corporations and entry.** A plant that serves market $(n, k)$ pays a fixed cost

$$F_{i,n} = f_k \, w_\ell \, f^d_{\ell n}, \tag{7}$$

in units of labor of the country where the plant is located, with $f_k > 0$ and $f^d_{\ell n} \geq 1$ raising the cost of serving distant markets. The cost is incurred once per market served, whatever the quantity sold, and stands for registration, distribution and marketing; it does not depend on who owns the plant. Because it is paid in labor, entry uses real resources, appears as wage income, and leaves the model homogeneous of degree one in wages. We abstract from a fixed cost of setting up a plant. Marginal costs are constant and every fixed cost is specific to a plant–market pair, so a group's problem separates across destinations: in each market $(n, k(g))$ the group chooses the subset $T_{g,n} \subseteq \mathcal{P}_g$ of its plants that serve $n$. A parent group can serve a market from a plant at home, from a plant in the destination, from a plant in a third country, or from several at once; a stand-alone firm decides only whether to serve $n$. A plant operates if it serves at least one market, so which affiliates exist, and where each of them sells, is an equilibrium outcome. This is a simplified version of the multi-plant location problem of Yang (2023).[5]

[4] A duty levied instead on the exporter's price $p_{i,n}/(1 + t_{\ell nk})$ yields the same prices whenever all the plants a group uses to serve $n$ face the same tariff, and differs from (6) only in how the wedge $t_{\ell nk}(\mu - 1)a_i d_{\ell n} q_{i,n}$ is divided between the treasury of $n$ and the group. When a group serves $n$ from origins that face different tariffs, a duty on the exporter's price makes the group's markup differ across its plants, and the property that a group's markup depends only on its total share (Section 1.2) is lost. Specification (6) preserves it.

[5] With a plant-level setup cost, opening a plant to serve one market would lower the cost of serving others from it, linking a group's choices across markets and turning its problem into a combinatorial one over locations and markets jointly (Tintelnot, 2017). Separability is what allows the entry problem to be solved market by market in Section 1.2.

**Profit repatriation.** Country $n$ owns a share $\theta_{gn} \geq 0$ of group $g$, with $\sum_n \theta_{gn} = 1$. Groups maximize their own profits whoever owns them: ownership determines where profits are spent, not how groups price or which markets they enter. The profit of a group net of fixed costs, $\Pi_g = \sum_n [\pi_{g,n} - \sum_{i \in T_{g,n}} F_{i,n}]$, where $\pi_{g,n}$ is its variable profit in market $(n, k(g))$ (Section 1.2), is paid to its owners in proportion to $\theta_{gn}$, wherever it is earned. Tariff revenue is rebated lump sum to the household of the country that collects it. Income in country $n$ is

$$X_n = w_n L_n + \sum_{g \in \mathcal{G}} \theta_{gn} \, \Pi_g + T_n, \qquad T_n = \sum_k \sum_{i \in \mathcal{A}_{nk}} t_{\ell(i)nk} \, a_i \, d_{\ell(i)n} \, q_{i,n}. \tag{8}$$

Profits are earned where groups sell and received where their owners reside, and $\theta$ is what separates the two. A natural benchmark assigns each group to the country of its ultimate parent, $\theta_{gn} = \mathbf{1}\{n = h(g)\}$; nothing in the definition of equilibrium requires this form. Since a group can always choose to serve no market, its net profit is non-negative in equilibrium.[6]

**Timing.** The model is static and all choices are made simultaneously. The pool of potential groups and all productivity draws are realized first. Given them, each group chooses which of its plants serve each market of its sector and how much each plant sells there, and households choose their consumption. Groups take wages, input prices and market sizes as given. When they choose quantities, they take into account their effect on the price index of the market; when they choose which plants operate, they take as given how competitive each market is, as summarized by the aggregate $A_{nk}$ of Section 1.2. Section 1.2 states this entry notion precisely, and Appendix B compares it with a two-stage game in which groups anticipate how their entry changes competition. In equilibrium, every choice is optimal given these objects, and the objects are those generated by the choices.

## 1.2 Market Structure

Fix a market $(n, k)$, take the set of plants that serve it as given until the last paragraph of this section, and suppress the market's indices: we write $\sigma$, $E$ and $P$ for $\sigma_k$, $E_{nk}$ and $P_{nk}$, $c_i$ for the delivered cost $c_{i,n}$, and $T_g$ for the plants of group $g$ that serve the market.

**Demand.** Let $A \equiv \sum_{i \in \mathcal{A}} p_i^{1-\sigma} = P^{1-\sigma}$. By (3), the revenue share of plant $i$ and the share of group $g$ are

$$s_i = \frac{p_i q_i}{E} = \frac{p_i^{1-\sigma}}{A}, \qquad S_g = \sum_{i \in T_g} s_i, \qquad \sum_g S_g = 1, \tag{9}$$

where $S_g = 0$ for a group that does not serve the market. The aggregate $A$ summarizes how competitive the market is: it is large when rivals charge low prices, and then any given price wins a small share.

**Competition.** Groups are the strategic agents. Each group chooses the quantities of all of its plants that serve the market, taking as given the quantities of other groups and the overall demand for the sector, which is the level of spending $E$ when $\eta = 1$. It maximizes its variable profit $\pi_g = \sum_{i \in T_g} (p_i - c_i) q_i$.

[6] Income includes the profits of the groups a country owns, wherever they produce and sell, so trade need not balance country by country. Adding (8) across countries and using (16) and (17) shows that world income equals world final spending. This is Walras' law, and it makes one labor-market condition redundant.

**Proposition 1** (Markups)**.** *Under quantity competition, the first-order conditions of group $g$ are $p_i = \mu(S_g)\, c_i$ for every $i \in T_g$, with*

$$\frac{1}{\mu(S)} = 1 - \frac{1-S}{\sigma} - \frac{S}{\eta}. \tag{10}$$

*Under price competition the markup is $\mu(S) = \varepsilon(S)/[\varepsilon(S)-1]$ with $\varepsilon(S) = \sigma - (\sigma - \eta)S$.*

Both are the markups of Atkeson and Burstein (2008), evaluated at the share of the group rather than of the plant, and both satisfy the following property whenever $\sigma > \eta$.

**Assumption 1.** $\mu$ is continuously differentiable and strictly increasing on $[0, 1)$, with $\mu(0) = \sigma/(\sigma - 1)$.

The existence and uniqueness results for a single market, for input prices and for wages use only Assumption 1. The results on entry and on profit income use the form of the markup under quantity competition, and we say so where they do. We take quantity competition as the baseline.[7]

**Cannibalization across multinational affiliates.** When a group expands one of its plants, the sector price index falls and every variety in the market loses sales, the group's other plants included. A single-plant firm ignores the losses it imposes on others; a group counts the losses it imposes on itself, and these grow with its share. A group with a negligible share competes only with rivals, at the within-sector elasticity $\sigma$. A group holding the whole market is the sector, and competes only with other sectors, at the elasticity $\eta$. Equation (10) weights the two by the group's share, so the markup rises from $\sigma/(\sigma-1)$ for a negligible group toward the monopoly markup. Two implications follow. First, all plants of a group charge the same markup in a given market, and differences in their costs show up in quantities rather than in markups. Second, market power is a property of the group: concentration should be measured at the level of groups, and a group charges more than its plants would charge as independent firms. Within a group, (9) and the common markup give

$$\frac{s_i}{S_g} = \frac{c_i^{1-\sigma}}{K_g}, \qquad K_g \equiv \sum_{i \in T_g} c_i^{1-\sigma}. \tag{11}$$

We call $K_g$ the *strength* of the group in the market: an index of its cost efficiency that rises with the number of plants serving the market and with how cheaply each delivers. Substituting $p_i = \mu(S_g)c_i$ into (9),

$$\psi(S_g) = \frac{K_g}{A}, \qquad \psi(S) \equiv S\,\mu(S)^{\sigma-1}. \tag{12}$$

Under Assumption 1 the map $\psi$ is strictly increasing, so $S_g = \psi^{-1}(K_g/A)$: the share of a group depends only on its strength relative to the competitiveness of the market. However many plants a group runs, and in however many countries, its position in the market is summarized by one number, $K_g$, and all it needs to know about its rivals is $A$. This holds under price competition as well, since only the form of $\mu$ changes. Given the active plants, the equilibrium of the market is the value of $A$ at which shares add up to one,

$$\sum_g \psi^{-1}\left(\frac{K_g}{A}\right) = 1. \tag{13}$$

[7] When $\eta = 1$ the markup in (10) is unbounded as $S \to 1$, so a market served by a single group has no equilibrium. Such a configuration is never an entry equilibrium as long as a second group can serve the market and the market's spending exceeds that group's fixed cost: facing a monopolist's unbounded price, the entrant would earn profits close to the market's spending. We therefore consider only configurations in which at least two groups are active in every market.

**Profits.** Variable profit is $\pi_g = E\, S_g\, m(S_g)$, where $m(S) \equiv 1 - 1/\mu(S)$ is the Lerner index. Under quantity competition (10) makes the Lerner index linear in the share, $m(S) = (1 + \kappa S)/\sigma$ with $\kappa \equiv \sigma/\eta - 1$, so that

$$\pi_g = \frac{E}{\sigma}(S_g + \kappa S_g^2). \tag{14}$$

Under price competition the Lerner index $1/\varepsilon(S)$ is not linear, and $\kappa = (\sigma - \eta)/\sigma$ is the slope of its first-order expansion. Profits rise more than proportionally with the share, because a larger group both sells more and earns more on each unit sold. The curvature $\kappa$ governs the ownership results of Section 1.4, and with constant markups $\kappa = 0$. Linearity is what makes the ownership decomposition of Section 1.4 exact rather than approximate, and it is the main reason we take quantity competition as the baseline.

**Entry.** A group's plant set matters for its profit only through its strength. Let $K(T) = \sum_{i \in T} c_i^{1-\sigma}$ and $\Phi(T) = \sum_{i \in T} F_{i,n}$ be the strength and the fixed cost of a set $T \subseteq \mathcal{P}_g$. Given $A$, serving the market with $T$ yields

$$V_g(T; A) = \Omega(K(T); A) - \Phi(T), \qquad \Omega(K; A) \equiv E\, \psi^{-1}(K/A)\, m(\psi^{-1}(K/A)). \tag{15}$$

A group with $|\mathcal{P}_g|$ potential plants chooses among $2^{|\mathcal{P}_g|}$ sets. Because $\Omega$ is strictly increasing in $K$, a set can be optimal only if no other set delivers at least as much strength at no greater fixed cost. These efficient sets form a chain, ordered by strength and by cost together, so the group's choice amounts to how far to climb a ladder that runs from small and cheap to large and expensive. A stand-alone firm has a ladder with a single rung. When a group's plants pay the same fixed cost to serve the market, the efficient sets are nested: each rung adds the plant with the next-lowest delivered cost, and the group's choice is a cutoff on its own list of plants, the within-group analogue of the cutoff in Gaubert and Itskhoki (2021). When fixed costs differ across its plants, the rungs need not be nested, but the ladder remains ordered by strength and cost.

**Definition 1** (Entry equilibrium in a market)**.** Given delivered costs $\{c_i\}$, fixed costs $\{F_{i,n}\}$ and spending $E$, an entry equilibrium is a collection of active sets $\{T_g\}$ and an aggregate $A$ such that (i) each $T_g$ maximizes $V_g(\cdot\,; A)$ over subsets of $\mathcal{P}_g$, and (ii) $A$ solves (13) with $K_g = K(T_g)$.

Groups take the competitiveness of the market as given when they decide which plants to operate, although they take into account their effect on $A$ when they set quantities. Each group thus evaluates entry at the prevailing degree of competition, as a free-entry condition does. The difference from a notion in which groups anticipate how their entry changes $A$ matters only for groups with large shares, and it vanishes as a group's share goes to zero. We adopt this notion because it turns each group's entry problem into the choice of one number given one number, which is what delivers uniqueness in Section 1.3. Appendix B discusses the stronger notion in which a group evaluates every deviation on the re-solved market.

## 1.3 General Equilibrium

**Market sizes.** Let $V_{i,n'} \equiv a_i\, d_{\ell(i)n'}\, q_{i,n'}$ be the variable cost of plant $i$ in serving country $n'$, exclusive of tariffs. By Proposition 1, $V_{i,n'} = r_{i,n'}/[\mu_{g(i)}(1 + t_{\ell(i)n'k})]$, where $r_{i,n'}$ is the plant's revenue in that market and $\mu_{g(i)}$ its group's markup there. Spending on market $(n, k)$ is final spending by households plus intermediate purchases by every plant located in $n$, wherever it sells:

$$E_{nk} = \frac{\beta_k P_{nk}^{1-\eta}}{P_n^{1-\eta}}\, X_n + \sum_{k' \in \mathcal{K}} \omega_{kk'}\, \nu_{k'} \sum_{n' \in \mathcal{N}} \sum_{i \in \mathcal{A}_{n'k'}:\, \ell(i) = n} V_{i,n'}. \tag{16}$$

Fixed costs are paid in labor only and generate no purchases of inputs.

**Labor market.** In each country, labor demanded for production and for serving markets equals the labor supplied:

$$w_n L_n = \sum_{k\in\mathcal{K}} (1-\nu_k) \sum_{n'\in\mathcal{N}} \sum_{i\in\mathcal{A}_{n'k}:\, \ell(i)=n} V_{i,n'} \;+\; \sum_{n'\in\mathcal{N}} \sum_{i:\, \ell(i)=n,\; i\in T_{g(i),n'}} F_{i,n'}. \tag{17}$$

**Definition 2** (Equilibrium)**.** An equilibrium consists of wages $\{w_n\}$, price indices $\{P_{nk}\}$, market sizes $\{E_{nk}\}$, incomes $\{X_n\}$, active sets $\{T_{g,n}\}$, and prices and quantities $\{p_{i,n}, q_{i,n}\}$ such that

(i) households maximize (1) given their income;

(ii) in every market, each group's quantities (prices, under price competition) maximize its variable profit given its rivals', so that prices satisfy Proposition 1;

(iii) in every market, quantities satisfy demand (3) and shares satisfy (13);

(iv) in every market, the active sets and the aggregate form an entry equilibrium (Definition 1), with fixed costs (7);

(v) costs (4)–(6) are evaluated at the price indices (2) that they generate;

(vi) market sizes satisfy (16) and incomes satisfy (8);

(vii) labor markets clear, (17).

All prices, costs, fixed costs and incomes are homogeneous of degree one in wages, and shares are homogeneous of degree zero. Since variable profits and fixed costs scale together, entry decisions are unchanged when all wages are scaled by a common factor, so wages are determined up to scale. By Walras' law one labor-market condition is implied by the others, and we set $w_1 = 1$.

**Existence and uniqueness.** Given wages, the model solves as a sequence of nested problems, and we establish existence and uniqueness layer by layer. The first two layers are single markets.

**Proposition 2** (One market)**.** *Under Assumption 1, for any active set containing at least two groups and any positive costs, the market has exactly one equilibrium.*

The proof uses (13): each $\psi^{-1}(K_g/A)$ falls as $A$ rises, so the sum of shares is a strictly decreasing function of a single number and equals one exactly once. Each group's profit is strictly concave in its own quantities, so the first-order conditions identify the unique best response.

Entry is harder. Each entry raises $A$ and lowers the value of entering for everyone else, and discrete games of this kind typically have several equilibria. The usual remedy, ranking entrants by cost and letting the most efficient enter first, requires that every entrant be a separate firm; once a group can add one of its own plants, an entry raises its share instead of lowering it and the ranking argument fails. Fixing an order of moves restores uniqueness only by selecting among equilibria. We instead separate the effect of a group's strength on its own payoff from the effect of its rivals. Write $\Omega_K$ and $\Omega_{KA}$ for the partial derivatives of $\Omega$.

**Theorem 1** (Entry)**.** *Consider quantity competition with $\eta = 1$ and $\sigma \geq 2$.*

*(a) $\Omega(K; A)$ is strictly concave in $K$ at every share.*

*(b) $\Omega_{KA} < 0$ whenever the group's share is at most one half.*

*(c) Suppose that no group's share exceeds one half at any strength profile between two candidate equilibria, and that no group is indifferent between two rungs of its ladder. Then a market has at most one entry equilibrium.*

Part (a) says that strength has diminishing returns for the group itself, and part (b) that strength is worth less in a more competitive market. Together they imply that each group's optimal strength falls as $A$ rises, while $A$ rises with every group's strength, so two equilibria with different aggregates cannot both exist. No order of entry is imposed and groups continue to internalize competition among their own plants. The hypothesis has economic content and can be checked: no single group holds more than half of a market. It is sufficient rather than necessary; at exactly one half, the expression whose sign gives (b) still equals $1/\sigma$. The proof covers quantity competition with $\eta = 1$. For price competition and for $\eta > 1$ we have verified numerically that the same one-half condition suffices, but we do not have a proof. Existence is a separate matter: entry is chosen in whole plants, so an exact equilibrium in pure strategies need not exist (Appendix B).

The remaining layers take active sets as given.

**Proposition 3** (Prices and incomes)**.** *Fix wages and active sets. (a) The price indices are the unique fixed point of* (2) *combined with* (4)–(6)*, and iteration converges to it at rate* $\max_k \nu_k$*. (b) Market sizes and incomes are the unique solution of the linear system* (8)*,* (16)*.*

**Proposition 4** (Existence of wages)**.** *Fix active sets and suppose every country hosts at least one active plant. Then there exist wages that clear every labor market.*

Uniqueness of wages is the last layer and the most delicate. Let $W_n(w)$ be the wage bill of country $n$ implied by (17) at wages $w$, and $M_{nj} \equiv \partial \ln W_n / \partial \ln w_j$. Homogeneity makes every row of $M$ sum to one. Labor in different countries is a gross substitute when $M_{nj} > 0$ for $n \neq j$: a rise in any other country's wage raises country $n$'s wage bill. The next lemma is the building block, and it holds without conditions.

**Lemma 1** (Cost pass-through)**.** *Fix active sets. The matrix* $\partial \ln P / \partial \ln w$*, with rows indexed by markets, is non-negative and its rows sum to one.*

A rise in one country's wage raises every price index in the world by a proportion between zero and one, whatever the markups and the input–output structure. At each step of the argument the response is a weighted average of responses one step down: a plant's cost averages wages and input prices, and a market's price index averages the cost changes of the groups in it, with weights that variable markups change but cannot make negative.

The model's solution algorithm adjusts wages by the damped tâtonnement

$$\Psi(w)_n = w_n^{1-\lambda} \big(W_n(w)/L_n\big)^{\lambda}, \qquad \lambda \in (0,1),$$

which raises a country's wage when its labor is in excess demand and whose fixed points, up to scale, are the equilibrium wage vectors. Let $d_H(w, w') = \max_n \ln(w_n/w'_n) - \min_n \ln(w_n/w'_n)$ denote Hilbert's projective metric, which is unchanged when either vector is rescaled.

**Theorem 2** (Wages)**.** *Fix active sets. Let* $D$ *be a set of wage vectors that is convex in logarithms, and suppose that on* $D$ *(i)* $M_{nj} > 0$ *for all* $n \neq j$*, and (ii)* $\lambda < 1/(1 - \min_n M_{nn})$*. Then there is* $\tau < 1$ *such that* $d_H(\Psi(w), \Psi(w')) \leq \tau\, d_H(w, w')$ *for all* $w, w' \in D$*. Hence* $D$ *contains at most one equilibrium wage vector up to scale, and if* $\Psi$ *maps* $D$ *into itself the algorithm converges to it from any starting point in* $D$*.*

Condition (ii) is not a numerical convenience. Labor demand is elastic, so $M_{nn}$ is negative and the undamped map ($\lambda = 1$) is not order-preserving; the model itself bounds the admissible step. Condition (i) is the substantive one, and the next result gives conditions under which it holds.

**Proposition 5** (Gross substitutes)**.** *Suppose $\nu_k = 0$ and $\eta = 1$, and that no group's share exceeds one half in any market under quantity competition. Then for $n \neq j$,*

$$M_{nj} \;\geq\; \bar{E}_n^{(j)} \equiv \sum_{i,n'} b_{i,n'}^n \, \frac{\partial \ln E_{n'k(i)}}{\partial \ln w_j}, \tag{18}$$

*where $b_{i,n'}^n$ is the share of country $n$'s wage bill paid by plant $i$ out of its variable costs in country $n'$. Labor is a gross substitute whenever $\bar{E}_n^{(j)} > 0$.*

The bound says that when production uses only local labor, a rise in a foreign wage can lower country $n$'s wage bill only by shrinking the markets in which its plants sell; every reallocation of sales between plants and between groups favors country $n$. The one-half condition is the same as in Theorem 1: it is what signs the effect of variable markups on factor demand. With intermediates, the same bound holds provided a plant's input prices do not rise by more than its group's average cost, a condition that is vacuous when $\nu_k = 0$.

Three limits of these results should be stated plainly. First, each layer is unique given the layers it takes as given, which does not make the joint equilibrium of wages and active sets unique: active sets change discretely with wages, and Theorem 2 holds them fixed. Second, an exact entry equilibrium in whole plants may fail to exist. Third, gross substitutes is not a theorem of the model. When head-office services are a factor input it can fail at an equilibrium (Appendix C), and with capability alone the demand term in (18) can still turn negative, because a country whose groups lose profits when a foreign wage rises spends less. That income effect is the ownership channel the model is built to study, and removing it would remove the object of interest. The conditions of Theorems 1 and 2 and of Proposition 5 are inequalities in objects the model computes, and we check them at the computed equilibrium in the quantification.

**Solution algorithm.** The structure above maps into nested loops. Given an aggregate $A$, each group's share solves the scalar equation (12), and the market solves (13) by safeguarded Newton steps on $\ln A$, which never leave a bracket that contains the unique root. With entry, each group's best response at a given $A$ is found by scanning its ladder of efficient sets, and $A$ is again found by bisection. Given wages and active sets, price indices are obtained by iterating on (4)–(2), which is a contraction, and market sizes and incomes by one linear solve. Wages are updated by the tâtonnement of Theorem 2, followed by Newton steps once close to the solution. An outer loop updates active sets: at the equilibrium wages for the current configuration, the groups with the largest gains from changing their sets do so, and the procedure repeats until no group wishes to move. When no exact fixed point in whole plants exists, the algorithm returns the configuration with the fewest groups wishing to move and reports how many do. Moving only the groups with the largest gains changes the path of the algorithm, not its fixed points, and does not amount to an order of entry.

### 1.4 Comparative Statics

**Incidence within a market.** Consider a market with given active sets and spending, and a change in delivered costs. Let $\hat{c}_g = \sum_{i \in T_g} (s_i / S_g)\, d \ln c_i$ be the average cost change of group $g$,

$\varepsilon_\mu(S) \equiv S\mu'(S)/\mu(S)$ the elasticity of the markup, and

$$\Lambda_g \equiv \frac{1}{1+(\sigma-1)\,\varepsilon_\mu(S_g)}, \qquad \varpi_g \equiv \frac{S_g\Lambda_g}{\sum_h S_h\Lambda_h}. \tag{19}$$

Under quantity competition with $\eta = 1$, $\Lambda_g = (1-S_g)/[1+(\sigma-2)S_g]$, which falls from one toward zero as the share rises.

**Proposition 6** (Incidence)**.** *Holding active sets and spending fixed,*

$$d\ln S_g = -(\sigma-1)\Lambda_g(\hat{c}_g - d\ln P), \qquad d\ln P = \sum_g \varpi_g\,\hat{c}_g. \tag{20}$$

*If only group $j$'s costs change, its price rises by the fraction $\rho_j = 1-(\sigma-1)\varepsilon_\mu(S_j)\Lambda_j(1-\varpi_j) \in (0,1]$ of the cost increase, and every rival $h$ gains share, $d\ln S_h = (\sigma-1)\Lambda_h\varpi_j\hat{c}_j$, and raises its price.*

Three features distinguish this from monopolistic competition, where $\Lambda_g = 1$, $\varpi_g = S_g$ and $\rho_j = 1$. Large groups absorb part of a cost increase in their markup, so a tariff on them is partly paid by the group: pass-through $\rho_j$ falls with the group's share. Since $\Lambda_g$ falls with size, $\varpi_g < S_g$ for the larger groups in a market, and the price index moves by less than their share would suggest. And rivals are not passive: a tariff on one group raises the shares, markups and prices of the others.

**Ownership and the level of profit income.** Consider a country $H$ with ownership shares $\theta_g \equiv \theta_{gH}$, and its profit income from a market, $\pi^H = \sum_g \theta_g\pi_g$.

**Proposition 7** (Ownership-weighted concentration)**.** *Under quantity competition,*

$$\frac{\pi^H}{E} = \frac{1}{\sigma}\Big[\bar{\theta} + \kappa\big(\bar{\theta}\cdot\mathrm{HHI} + \mathrm{Cov}_S(\theta,S)\big)\Big], \tag{21}$$

*where $\bar{\theta} = \sum_g S_g\theta_g$ is the sales-weighted ownership share,* $\mathrm{HHI} = \sum_g S_g^2$*, and* $\mathrm{Cov}_S(\theta,S) = \sum_g \theta_g S_g^2 - \bar{\theta}\cdot\mathrm{HHI}$*. A calculation that applies the aggregate ownership share to total profit, $\bar{\theta}\sum_g \pi_g$, misses exactly $(E\kappa/\sigma)\,\mathrm{Cov}_S(\theta,S)$.*

The first term is what a model with constant markups delivers: ownership times revenue over $\sigma$. The second is an ownership-weighted Herfindahl index. Its component $\bar{\theta}\cdot\mathrm{HHI}$ can be built from an aggregate ownership share and a published concentration index. The covariance between ownership and size cannot: it requires knowing which groups a country owns and how large each one is. Because profits are convex in shares, what matters for a country's profit income is not what fraction of firms it owns but what fraction of the large ones. The decomposition is an identity. It is exact under quantity competition, where the Lerner index is linear; under price competition it is the leading term of an expansion.

**Corollary 1** (Constant markups)**.** *If $\kappa = 0$, a country's profit income from a market equals its aggregate ownership share times total profit, whatever the distribution of ownership across groups.*

The corollary concerns the level of income given $\bar{\theta}$. It does not say that the distribution of ownership is irrelevant for how income responds to policy, which is the subject of the next result.

**Ownership and the response of profit income.** Policy acts through changes in profit income, not through its level, and the two are governed by different statistics.

**Proposition 8** (Response of owned profits)**.** *Under quantity competition with $\eta = 1$, holding active sets and spending fixed, a change in the delivered costs of group $j$ alone changes country $H$'s profit income by*

$$\frac{d\pi^H}{E} = \frac{\sigma - 1}{\sigma}\, \varpi_j\, \hat{c}_j \sum_{h \neq j} S_h \Lambda_h \Big[\theta_h(1 + 2\kappa S_h) - \theta_j(1 + 2\kappa S_j)\Big]. \tag{22}$$

A cost increase on group $j$, such as a tariff on the plants that serve the market, reallocates sales from $j$ to its rivals in proportion to $S_h\Lambda_h$, and each unit of share is worth $\theta(1 + 2\kappa S)$ to country $H$. The owner gains when the rivals it owns are worth more to it at the margin than the group being taxed. Two differences from Proposition 7 matter. The weights are $S_h\Lambda_h$ rather than $S_h^2$, and $\Lambda_h$ falls with size, so large groups count for less in the response than in the level. And the response depends on the ownership of the taxed group relative to its rivals, not on the covariance of ownership with size. The ownership-weighted Herfindahl index measures the stake a country has in a market; it is not, by itself, a sufficient statistic for how that stake responds to a tariff, and in general equilibrium the response also runs through wages, entry and market sizes.

**Entry.** The proof of Theorem 1 delivers monotone comparative statics for each group's choice.

**Corollary 2** (Entry responses)**.** *Under the conditions of Theorem 1, the strength a group chooses is non-increasing in the aggregate $A$ and non-decreasing in spending $E$, holding the other fixed.*

A more competitive market leads each group to climb fewer rungs of its ladder, and a larger market to climb more. Since every rung is a set of plants, these responses determine which affiliates operate, and through (16) and (17) they feed back into market sizes and wages.

**Wages and prices.** Lemma 1 is also a comparative statics result. A rise in country $j$'s wage raises every price index, in every country and sector, by a proportion between zero and one, and an equal proportional rise in all wages raises every price index in the same proportion. A wage change therefore propagates through the input–output structure and through markups, but neither can reverse its sign or amplify it beyond one for one. How the wage change is shared between consumers in different countries depends on where the plants that use country $j$'s labor sell, and how it is shared between workers and owners depends on who owns the groups that operate those plants.

## References


Antràs, Pol. "Firms, Contracts, and Trade Structure." *Quarterly Journal of Economics* 118(4): 1375–1418, 2003.

Antràs, Pol, and Elhanan Helpman. "Global Sourcing." *Journal of Political Economy* 112(3): 552–580, 2004.

Atkeson, Andrew, and Ariel Burstein. "Pricing-to-Market, Trade Costs, and International Relative Prices." *American Economic Review* 98(5): 1998–2031, 2008.

Birkhoff, Garrett. "Extensions of Jentzsch's Theorem." *Transactions of the American Mathematical Society* 85(1): 219–227, 1957.

Caliendo, Lorenzo, and Fernando Parro. "Estimates of the Trade and Welfare Effects of NAFTA." *Review of Economic Studies* 82(1): 1–44, 2015.

Gaubert, Cecile, and Oleg Itskhoki. "Granular Comparative Advantage." *Journal of Political Economy* 129(3): 871–939, 2021.

Head, Keith, and Thierry Mayer. "Brands in Motion: How Frictions Shape Multinational Production." *American Economic Review* 109(9): 3073–3124, 2019.

Mas-Colell, Andreu, Michael D. Whinston, and Jerry R. Green. *Microeconomic Theory*. Oxford University Press, 1995.

Milgrom, Paul, and Chris Shannon. "Monotone Comparative Statics." *Econometrica* 62(1): 157–180, 1994.

Ramondo, Natalia, and Andrés Rodríguez-Clare. "Trade, Multinational Production, and the Gains from Openness." *Journal of Political Economy* 121(2): 273–322, 2013.

Tintelnot, Felix. "Global Production with Export Platforms." *Quarterly Journal of Economics* 132(1): 157–209, 2017.

Yang, Chenying. "Location Choices of Multi-plant Oligopolists: Theory and Evidence from the Cement Industry." Working paper, 2023.

# Appendix

## A The Market Game

**A.1 Markups.** Fix a market. With $\eta$ the upper-tier elasticity, write spending as $E = D\,P^{1-\eta}$, with the shifter $D$ taken as given by firms; $\eta = 1$ gives $E = D$. Let $\rho = (\sigma-1)/\sigma$ and $Q = (\sum_i q_i^\rho)^{1/\rho}$. Inverse demand is $p_i = D^{1/\eta} Q^{1/\sigma - 1/\eta} q_i^{-1/\sigma}$, so group $g$'s revenue is $R_g = D^{1/\eta} Q^{1/\sigma-1/\eta} \sum_{j\in T_g} q_j^\rho$. Using $\partial \ln Q/\partial q_i = q_i^{\rho-1}/Q^\rho$ and $s_i = q_i^\rho/Q^\rho$,

$$\frac{\partial R_g}{\partial q_i} = p_i\Big[\rho + \Big(\frac{1}{\sigma} - \frac{1}{\eta}\Big) S_g\Big] = p_i\Big[1 - \frac{1-S_g}{\sigma} - \frac{S_g}{\eta}\Big].$$

Setting marginal revenue equal to $c_i$ gives (10); the bracket is common to all $i \in T_g$ and depends only on $S_g$. Under price competition, demand is $q_i = p_i^{-\sigma} P^{\sigma-\eta} D$, so $\partial \ln q_j/\partial \ln p_i = -\sigma \mathbf{1}\{i = j\} + (\sigma-\eta) s_i$. The first-order condition for $p_i$ is $q_i - \sigma(p_i - c_i) q_i/p_i + (\sigma-\eta)(s_i/p_i)\sum_{j\in T_g}(p_j - c_j) q_j = 0$. Trying $p_j = \mu c_j$ for all $j \in T_g$, so that $(p_j - c_j) q_j = (1 - 1/\mu) r_j$ and $\sum_{j\in T_g} r_j = S_g E$, it reduces to $(1 - 1/\mu)\big[\sigma - (\sigma-\eta) S_g\big] = 1$, which is the stated markup.

**A.2 Proof of Proposition 2.** *The share map.* $\psi(S) = S\mu(S)^{\sigma-1}$ is the product of $S$ and of $\mu(S)^{\sigma-1}$, both positive and strictly increasing on $(0,1)$ under Assumption 1, and $\psi(0) = 0$; so $\psi$ has a continuous, strictly increasing inverse. *Best responses.* Given rivals' quantities, group $g$'s profit is $D^{1/\eta} X_g (X_g + B_g)^e - \sum_{i\in T_g} c_i q_i$, where $X_g = \sum_{i\in T_g} q_i^\rho$, $B_g$ is the same index for rivals, and $e = (\eta-\sigma)/[\eta(\sigma-1)] \in [-1, 0)$. The function $X \mapsto X(X+B)^e$ is strictly increasing and strictly concave for $B > 0$: its derivatives are $(X+B)^{e-1}[(1+e)X + B] > 0$ and $e(X+B)^{e-2}[(1+e)X + 2B] < 0$. The index $X_g$ is strictly concave in $(q_i)_{i\in T_g}$. A strictly increasing, strictly concave function of a strictly concave function is strictly concave, and cost is linear, so profit is strictly concave in the group's own quantities. Since $\partial X_g/\partial q_i \to \infty$ as $q_i \to 0$, the optimum is interior and is the unique solution of the first-order conditions. *The aggregate.* By Proposition 1 and (12), an equilibrium is an $A$ solving (13), and conversely any such $A$ generates prices $p_i = \mu(S_g) c_i$ that satisfy every group's first-order conditions and are consistent with $A$, since $\sum_i p_i^{1-\sigma} = \sum_g \mu(S_g)^{1-\sigma} K_g = A \sum_g S_g = A$. The left side of (13), $\mathcal{S}(A)$, is continuous and strictly decreasing. As $A \to \infty$ every share tends to zero. Let $\bar\psi = \lim_{S\to 1} \psi(S)$, infinite when $\eta = 1$; as $A$ falls toward $\max_g K_g/\bar\psi$ the largest group's share tends to one while every other share stays positive, so $\mathcal{S}$ exceeds one. Hence $\mathcal{S}(A) = 1$ has exactly one solution, and every other variable is a function of it. □

## B Entry

**B.1 Two elasticities.** Let $G(S) \equiv \pi'(S)/\psi'(S)$, where $\pi(S) = ES\,m(S)$, and define $\varepsilon_G = SG'/G$ and $\varepsilon_\psi = S\psi'/\psi$. Differentiating $\psi(S) = K/A$ gives $\partial S/\partial K = 1/(A\psi')$ and $\partial S/\partial A = -\psi/(A\psi')$. Hence $\Omega_K = G(S)/A > 0$, $\Omega_{KK} = G'(S)/(A^2\psi'(S))$, and

$$\Omega_{KA} = -\frac{1}{A^2}\Big[G'(S)\frac{\psi(S)}{\psi'(S)} + G(S)\Big] = -\frac{G(S)}{A^2 \varepsilon_\psi(S)}\big[\varepsilon_G(S) + \varepsilon_\psi(S)\big].$$

So $\Omega$ is strictly concave in $K$ if and only if $\varepsilon_G < 0$, and $\Omega_{KA} < 0$ if and only if $\varepsilon_G + \varepsilon_\psi > 0$.

**B.2 Proof of Theorem 1(a)–(b).** With $\eta = 1$, $\mu(S) = \sigma/[(\sigma-1)(1-S)]$. Set $E = 1$, $a \equiv 2(\sigma-1)$ and $b \equiv \sigma - 2 \geq 0$. Then $\pi'(S) = (1+aS)/\sigma$ and $\psi'(S) = C(1-S)^{-\sigma}(1+bS)$ with $C = [\sigma/(\sigma-1)]^{\sigma-1}$, so

$$\varepsilon_G(S) = S\,f(S), \qquad f(S) = \frac{a}{1+aS} - \frac{b}{1+bS} - \frac{\sigma}{1-S}.$$

At $S = 0$, $f(0) = a - b - \sigma = 0$. Since $a > b \geq 0$, $a/(1+aS) > b/(1+bS) \geq 0$, so $f'(S) = -a^2/(1+aS)^2 + b^2/(1+bS)^2 - \sigma/(1-S)^2 < 0$. Hence $f < 0$ and $\varepsilon_G < 0$ on $(0,1)$, which is (a). Next, $\varepsilon_\psi = (1+bS)/(1-S)$, and using $1 + bS - \sigma S = 1 - 2S$,

$$\varepsilon_G + \varepsilon_\psi = \frac{aS}{1+aS} - \frac{bS}{1+bS} + \frac{1-2S}{1-S}.$$

The map $x \mapsto x/(1+x)$ is increasing and $a > b$, so the first two terms sum to a positive number; the third is non-negative when $S \leq \frac{1}{2}$. This is (b). At $S = \frac{1}{2}$ the expression equals $(\sigma-1)/\sigma - (\sigma-2)/\sigma = 1/\sigma$. □

**B.3 Proof of Theorem 1(c).** *Step 1: the ladder.* Since $\Omega_K > 0$, a set $T$ is optimal only if no $T'$ has $K(T') \geq K(T)$ and $\Phi(T') \leq \Phi(T)$ with one inequality strict. The remaining efficient sets can be ordered so that $K$ and $\Phi$ rise together. *Step 2: single crossing.* For two rungs with $K_1 < K_2$, let $\Delta(A) = \int_{K_1}^{K_2} \Omega_K(K; A)\, dK - [\Phi_2 - \Phi_1]$ be the net gain from the larger one. Under the hypothesis, $\Omega_{KA} < 0$ on $[K_1, K_2]$, so $\Delta$ is strictly decreasing in $A$. Hence if the larger rung is preferred at some $A'$, it is preferred at every $A < A'$, and with no ties the optimal strength $K_g^*(A)$ is non-increasing in $A$ (Milgrom and Shannon, 1994). *Step 3: the aggregate.* Given strengths, the aggregate $A(\{K_g\})$ solving (13) is strictly increasing in each $K_g$: raising $K_j$ at a given $A$ raises the left side, which is strictly decreasing in $A$. *Step 4.* Suppose $(A, \{K_g\})$ and $(A', \{K_g'\})$ are equilibria with $A < A'$. By Step 2, $K_g' = K_g^*(A') \leq K_g^*(A) = K_g$ for all $g$, so by Step 3, $A' = A(\{K_g'\}) \leq A(\{K_g\}) = A$, a contradiction. Equilibria with the same aggregate have the same strengths, and with no ties the same active sets. □

**B.4 The notion of entry equilibrium and existence.** Definition 1 evaluates every alternative set at the prevailing aggregate $A$. In a two-stage game in which the market is re-solved after a deviation, a group that adds a plant anticipates that $A$ rises, which lowers the value of the deviation; the two notions coincide for groups with negligible shares and can differ for large ones. Theorem 1 applies to the notion of Definition 1. For the two-stage notion we have only numerical evidence, from markets small enough to enumerate every configuration and re-solve the market after every deviation. Existence in pure strategies is not guaranteed under either notion: a group's choice is discrete, so $K_g^*(A)$ jumps, and (13) may have no solution with each group on a rung. The algorithm of Section 1.3 then returns the configuration with the fewest groups wishing to deviate.

## C General Equilibrium

**C.1 Proof of Proposition 3.** *(a)* Let $T$ map log price indices $\ln P$ into the log price indices generated by the costs (4)–(6) that $P$ implies, holding wages and active sets fixed. Multiplying every cost in a market by a constant leaves shares and markups unchanged and multiplies every price by that constant, so each market's price index is homogeneous of degree one in the costs entering it; by Lemma 1's second step it is also non-decreasing in each of them. In (4), $\ln a_i$ depends on $\ln P$ through $\nu_k \sum_{k'} \omega_{k'k} \ln P_{\ell k'}$, with non-negative weights summing to $\nu_k$. Hence $T$ is monotone and $T(\ln P + x\mathbf{1}) \leq T(\ln P) + (\max_k \nu_k)\, x\mathbf{1}$ for $x \geq 0$, so by Blackwell's sufficient conditions $T$ is a contraction in the sup norm with modulus $\max_k \nu_k < 1$. *(b)* Given shares and markups, the second term of (16) is linear in market sizes, $E = \mathcal{E}X + BE$, where $B \geq 0$ and the column of $B$ for market $(n', k')$ sums to $\nu_{k'} \sum_i s_i/[\mu(1+t)] \leq \nu_{k'} < 1$. So $\rho(B) < 1$ and $E = (I - B)^{-1}\mathcal{E}X$. Variable profits and tariff revenue are linear in $E$, so (8) reads $X = wL - \theta'\Phi + \mathcal{M}X$ with $\mathcal{M} \geq 0$. Summing over countries, $\mathbf{1}'\mathcal{M}X$ equals world spending minus intermediate purchases minus the variable wage bill, that is world final spending minus the variable wage bill; since every unit of spending generates a

positive variable wage bill when $\nu_k < 1$, each column of $\mathcal{M}$ sums to less than one, $\rho(\mathcal{M}) < 1$, and $X = (I - \mathcal{M})^{-1}(wL - \theta'\Phi)$ is the unique solution. □

**C.2 Proof of Proposition 4.** Let $z_n(w) = W_n(w)/w_n - L_n$ be excess demand for labor. By Propositions 2–3, $z$ is continuous on the interior of the wage space, as a composition of continuous maps; it is homogeneous of degree zero; it satisfies Walras' law $\sum_n w_n z_n(w) = 0$, which follows from adding (8) across countries and using (16); and it is bounded below by $-L$. For the boundary condition, let $w_n \to 0$ with other wages bounded away from zero. Spending in every market stays bounded away from zero, since incomes are bounded below by $w_j L_j - \sum_g \theta_{gj}\Phi_g$, which we assume positive (fixed costs small relative to labor income). Delivered costs of plants located in $n$ fall at least in proportion to $w_n^{1-\nu_k}$, and their variable cost bills are $Es_i/[\mu(1+t)]$. If the group's share stays below one, the markup is bounded and the bill stays bounded away from zero. If the share tends to one (possible only when $\eta = 1$), (12) implies that $1 - S_g$ falls like $c_i^{(\sigma-1)/\sigma}$, so the bill falls like $w_n^{(1-\nu_k)(\sigma-1)/\sigma}$, more slowly than $w_n$. Either way $z_n = W_n/w_n - L_n \to \infty$. The standard existence result for excess demand functions (Mas-Colell, Whinston, and Green, 1995, Proposition 17.C.1) applies. □

**C.3 Proof of Lemma 1.** Hats denote log changes in response to a change in wages. *Costs.* Trade costs and tariffs are fixed, so by (4) and (6), $\hat{c}_i = \hat{a}_i = (1-\nu_k)\hat{w}_\ell + \nu_k \sum_{k'} \omega_{k'k}\hat{P}_{\ell k'}$, with non-negative coefficients summing to one. *Markets.* Differentiating $S_g = \mu(S_g)^{1-\sigma}K_g/A$, with $\hat{K}_g = (1-\sigma)\hat{c}_g$ by (11) and $\hat{A} = (1-\sigma)\hat{P}$, gives $\hat{S}_g = -(\sigma-1)\Lambda_g(\hat{c}_g - \hat{P})$. Shares sum to one, so $\sum_g S_g\hat{S}_g = 0$, which gives $\hat{P} = \sum_g \varpi_g \hat{c}_g$ with weights $\varpi_g > 0$ summing to one. *Stacking.* Substituting costs into markets gives $\hat{P} = \Gamma\hat{w} + \mathcal{N}\hat{P}$ with $\Gamma, \mathcal{N} \geq 0$, $(\Gamma + \mathcal{N})\mathbf{1} = \mathbf{1}$, and the rows of $\mathcal{N}$ summing to $\nu_k < 1$. Hence $\rho(\mathcal{N}) < 1$, $(I - \mathcal{N})^{-1} = \sum_{s\geq 0}\mathcal{N}^s \geq 0$, and $\partial \ln P/\partial \ln w = (I - \mathcal{N})^{-1}\Gamma \geq 0$ with row sums $(I - \mathcal{N})^{-1}(\mathbf{1} - \nu) = \mathbf{1}$. □

**C.4 A decomposition of factor payments.** Let $V_i$ be plant $i$'s variable cost in a market $(n', k)$, with tariffs fixed. Since $V_i = r_i/[\mu_g(1+t)]$, $\hat{V}_i = \hat{E} + \hat{s}_i - \hat{\mu}_g$, with $\hat{s}_i = (1-\sigma)(\hat{c}_i + \hat{\mu}_g - \hat{P})$ and $\hat{\mu}_g = \varepsilon_\mu \hat{S}_g$. Using C.3 and $(\sigma - 1) = (\sigma-1)\Lambda_g[1 + (\sigma-1)\varepsilon_\mu]$,

$$\hat{V}_i = \hat{E} - (\sigma-1)(\hat{c}_i - \hat{c}_g) - (1-\varepsilon_\mu)(\sigma-1)\Lambda_g(\hat{c}_g - \hat{P}). \tag{A.1}$$

The first term is the growth of the market, the second the reallocation of sales within the group, the third the reallocation between the group and its rivals. Its coefficient is non-negative if and only if $\varepsilon_\mu \leq 1$, which under quantity competition with $\eta = 1$, where $\varepsilon_\mu = S/(1-S)$, is exactly $S_g \leq \frac{1}{2}$.

**C.5 Proof of Proposition 5.** Raise $w_j$ alone, $j \neq n$. With $\nu_k = 0$ and head-office services a capability, the whole variable labor bill of a plant is paid in its location, so the plants paying country $n$ are located in $n$, and their costs do not respond: $\hat{c}_i = 0$. Fixed costs (7) are paid at $w_n$ and do not respond either. By Lemma 1, $\hat{c}_g, \hat{P} \in [0, 1]$. Let $\chi = (1-\varepsilon_\mu)(\sigma-1)\Lambda_g$. The one-half condition and $\Lambda_g \leq 1$ give $0 \leq \chi \leq \sigma - 1$, so the last two terms of (A.1) equal $(\sigma-1)\hat{c}_g - \chi(\hat{c}_g - \hat{P})$. If $\hat{c}_g \geq \hat{P}$ this is at least $(\sigma-1)\hat{P}$; otherwise both parts are non-negative. In either case it is at least $(\sigma-1)\min\{\hat{c}_g, \hat{P}\} \geq 0$, so $\hat{V}_i \geq \hat{E}$. Weighting by $b^n_{i,n'}$ and adding the fixed-cost part, whose response is zero, gives (18). □

**C.6 Gross substitutes and uniqueness.** The following classical argument is included for completeness. *If $M_{nj} > 0$ for $n \neq j$ at every point of a set $D$ that is convex in logarithms, $D$ contains at most one equilibrium up to scale.* Let $w, w^* \in D$ be non-proportional equilibria, scale $w^*$ so that $w^* \leq w$ with equality at some $n_0$, and move from $w^*$ to $w$ along the segment in logarithms. Every wage other than $w_{n_0}$ weakly rises and at least one strictly rises, so $W_{n_0}$ strictly rises while $w_{n_0}$ does not, and $z_{n_0}(w) > z_{n_0}(w^*) = 0$, a contradiction.

**C.7 Proof of Theorem 2.** In logarithms $\Psi$ becomes a map with Jacobian $(1-\lambda)I + \lambda M$, whose rows sum to one. Its off-diagonal entries are positive by (i) and its diagonal entries are positive by (ii), so it is a strictly positive stochastic matrix at every point of $D$. In logarithms $d_H$ is the oscillation seminorm $\operatorname{osc}(x) = \max_n x_n - \min_n x_n$. By the mean value theorem in integral form, the difference between the images of two points is $\bar{A}$ times the difference between the points, where $\bar{A}$ is the average of the Jacobian along the segment joining them, which stays in $D$. The contraction coefficient of a stochastic matrix in the oscillation seminorm, $\delta(A) = \frac{1}{2}\max_{i,j} \|A_{i\cdot} - A_{j\cdot}\|_1$, is convex in $A$, so $\delta(\bar{A}) \leq \sup_D \delta$, and for a strictly positive matrix $\delta(A) \leq \tanh(\Delta(A)/4) < 1$, where $\Delta(A) = \max_{i,j,k,l} \ln(A_{ik}A_{jl}/A_{jk}A_{il})$ (Birkhoff, 1957). Taking $\tau = \sup_D \tanh(\Delta/4)$ gives the contraction. Two fixed points in $D$ would contradict it; convergence additionally needs the iterates to stay in $D$. □

**C.8 Head-office services as a factor input.** Suppose instead that the labor term in (4) is $(w_h^{\alpha_k} w_\ell^{1-\alpha_k})^{1-\nu_k}$, so that country $h$ is paid the share $\alpha_k$ of the labor bill of every plant its groups run. All plants of a group report to the same headquarters, so country $h$ receives the share $\alpha_k$ of the group's whole bill in the market, and the within-group term of (A.1) averages to zero over the group. Consider a country $n$ whose labor income consists only of head-office payments from groups whose plants are all located in a single country $j$, with $\nu_k = 0$. Raising $w_j$ raises those groups' costs by $\hat{c}_g = 1 - \alpha_k$, while the price indices of the markets they serve rise only in proportion to the groups' incidence weights. By (A.1), $M_{nj}$ is then a weighted sum over these groups of $\hat{E} - (1-\varepsilon_\mu)(\sigma-1)\Lambda_g(\hat{c}_g - \hat{P})$. When the groups are small, $\Lambda_g \approx 1$, $\varepsilon_\mu \approx 0$ and $\hat{c}_g - \hat{P} \approx 1 - \alpha_k$, so $M_{nj}$ is negative whenever $(\sigma-1)(1-\alpha_k)$ exceeds the growth of the markets they serve, although no group is near one half of any market. The failure is worst when $\alpha_k$ is small: a large head-office share ties the group's cost to its own country's wage and insulates it from $w_j$. What breaks gross substitutes is a country whose income is concentrated on firms whose costs are set by a single foreign country.

## D Comparative Statics

**D.1 Proof of Proposition 6.** The first two equations are the market step of C.3, with the active sets and $E$ fixed. If only $j$'s costs change, $d\ln P = \varpi_j \hat{c}_j$, so $d\ln S_j = -(\sigma-1)\Lambda_j(1-\varpi_j)\hat{c}_j$ and, for $h \neq j$, $d\ln S_h = (\sigma-1)\Lambda_h \varpi_j \hat{c}_j$. Prices are $p_i = \mu(S_g)c_i$, so $d\ln p_i = \hat{c}_i + \varepsilon_\mu(S_g)\, d\ln S_g$, which gives $\rho_j$ for group $j$ and a positive price change for every rival. Since $(\sigma-1)\varepsilon_\mu\Lambda_j < 1$ and $1-\varpi_j \in [0,1)$, $\rho_j \in (0,1]$. Under quantity competition with $\eta = 1$, $\varepsilon_\mu = S/(1-S)$, which gives the stated $\Lambda_g$. □

**D.2 Proof of Proposition 7.** With $m(S) = (1+\kappa S)/\sigma$, $\pi^H/E = \sigma^{-1}\sum_g \theta_g S_g(1+\kappa S_g) = \sigma^{-1}[\bar{\theta} + \kappa \sum_g \theta_g S_g^2]$, and $\sum_g \theta_g S_g^2 = \bar{\theta}\cdot\mathrm{HHI} + \mathrm{Cov}_S(\theta, S)$ by definition. Total profit is the case $\theta_g \equiv 1$, $\sum_g \pi_g/E = (1+\kappa\,\mathrm{HHI})/\sigma$, using $\sum_g S_g = 1$. Subtracting $\bar{\theta}\sum_g \pi_g$ from $\pi^H$ leaves $(E\kappa/\sigma)\mathrm{Cov}_S(\theta, S)$. Corollary 1 is the case $\kappa = 0$. □

**D.3 Proof of Proposition 8.** By (14), $d\pi_g = (E/\sigma)(1 + 2\kappa S_g)\, dS_g$. Let $Z = \sum_h S_h \Lambda_h$, so $\varpi_j = S_j \Lambda_j / Z$. From D.1, $dS_j = -(\sigma - 1) S_j \Lambda_j (1 - \varpi_j) \hat{c}_j = -(\sigma - 1) \varpi_j \hat{c}_j \sum_{h \neq j} S_h \Lambda_h$ and $dS_h = (\sigma - 1) \varpi_j S_h \Lambda_h \hat{c}_j$ for $h \neq j$. Substituting into $d\pi^H = \sum_g \theta_g d\pi_g$ and collecting terms gives (22). □

**D.4 Proof of Corollary 2.** Monotonicity in $A$ is Step 2 of B.3. For spending, $\Omega(K; A) = E\, \tilde{\Omega}(K; A)$ with $\tilde{\Omega}$ strictly increasing in $K$, so for two rungs with $K_1 < K_2$ the net gain $E[\tilde{\Omega}(K_2; A) - \tilde{\Omega}(K_1; A)] - [\Phi_2 - \Phi_1]$ is strictly increasing in $E$, and the larger rung, once preferred, remains preferred as $E$ rises. □